1. Definition

The Travelling Salesman Problem (TSP) is a problem in which a salesman must visit every city exactly once, return to the starting city, and minimize the total travel distance.

2. Algorithm

1. Start from the first city.
2. Mark the current city as visited.
3. Find the nearest unvisited city.
4. Move to that city and add the distance to the total cost.
5. Repeat steps 2–4 until all cities are visited.
6. Return to the starting city.
7. Display the route and total distance.

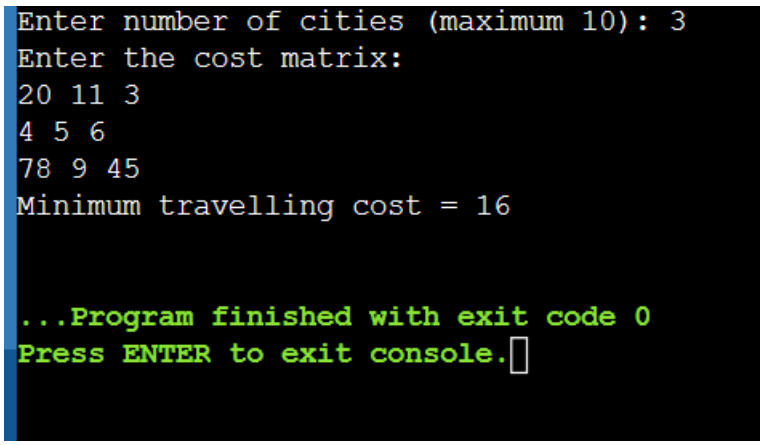
3. Time Complexity

Time Complexity: O(n^2)

4. Space Complexity

Space Complexity:O(n^2)

5. Output



6. Conclusion

The Travelling Salesman Problem program finds a tour by visiting all cities and returning to the starting city. It uses the greedy nearest-neighbour approach to select the nearest unvisited city at each step. This method is simple and efficient, but it does not always guarantee the minimum possible total distance.